Arya Mukherjee

07389064590 | arya.mukherjee01@gmail.com | aryam.dev | linkedin.com/in/aryamukherjee

Education

University of Bath *Sept 2019 - July 2024*

- MEng (Hons) Mechanical Engineering
- 2:1 Upper Second-Class Honours

Littleover Community School *2014 - 2019*

- A-Levels - Mathematics: A*, Physics: A, Computer Science: A
- 10 A* to As in GCSEs

Experience

Associate Software Engineer | SITA *Sep 2024 - Present*

- Designed and developed a distributed Python system for large-scale codebase analysis and summarisation, greatly reducing the onboarding time for new team members across the company - improving time-to-productivity by over 2 weeks.
- Architected and implemented a task parallelisation scheme for the summariser using Celery, distributing compute across 10 systems, improving inference speed by 500% and reducing monthly operational costs by 60%.
- Led a team to develop a product to simplify and centralise a partner team's testing framework, by building an automated soak testing system with FastAPI, React, Nginx and Kubectl as the interface with the control plane.
- Wrote Azure pipelines and Ansible playbooks to automate the building of the summariser and soak tester images, the configuration and hardening of the host linux VMs, and the deployment onto said VMs with services like Podman, Nginx and ELK stack monitoring.
- Written unit, regression, and end-to-end test suites in Python with pytest.
- Team lead and Scrum Master for 8 graduate developers, overseeing work delegation, mentoring and team coordination.
- **Technical Skills:** *Python, Docker, Linux, Git, Azure Pipelines, Ansible, Neo4j, Kubernetes, Playwright, pytest, basedpyright, uv, Ollama, OpenAI, Azure Cloud, Agile, Tree-sitter*

Mechanical Engineer | PRAX Lindsey Oil Refinery *July 2021 - July 2022*

- Assisted in scheduling a £40m project, processing contractor data in Oracle Primavera to ensure accurate and timely project execution.
- Automated QA data workflows with Python and Pandas, reducing spreadsheet processing times by 2000%, improving reporting speed and reliability for key safety and compliance checks.
- Processed site-wide operational data into interactive dashboards, using SAP and PowerBI.
- **Technical Skills:** *Python, Pandas, NumPy, Re, PowerBI, PowerQuery, Oracle Primavera, SAP*

Projects

Physics Informed Neural Network Modeller (Master's Thesis) *Jan 2024 - Apr 2024*

- Developed a Physics Informed Neural Network in PyTorch to model 3D transient heat system PDEs in gas turbines, achieving highly accurate simulations which generated 10x faster than traditional numerical methods.
- Accelerated and improved training by over 100x with CUDA parallelisation, the L-BFGS optimiser, gradient clipping and mixed precision training.
- Conducted various network hyperparameter space studies to determine the optimal architecture, activation function and weight initialisations - to expedite network convergence whilst avoiding the vanishing gradient problem.
- **Tools used:** *Deep Learning, Neural Networks, PyTorch, PDEs, NVIDIA CUDA, Plotly, Code Profiling, Jupyter Notebooks*

Neural Network Options Pricer (In Progress) *Apr 2026 - Present*

- Scoping and designing a low-latency C++ options pricer using the Black–Scholes model, with a focus on concurrent architecture and reliability. The pricer will use a Physics Informed Neural Network, with a custom-built C++ neural network library.
- **Tools used:** *C++ 23, Multithreading, Black–Scholes-Merton, Backtesting*

Various Numerical Modelling and Simulation Projects *Sep 2020 - Dec 2023*

- Created multiple numerical solvers in MATLAB for partial differential equations in heat transfer, orbital mechanics and drug absorption.
- Utilised computer vision to extract graph data from various NASA publications, to model 3D heat profiles of a space shuttle upon re-entry.
- Applied vectorisation and multi-threading to accelerate calculations and improve efficiency.
- Enhanced interactivity by creating GUIs for the numerical solvers, allowing users to run simulations and visualise results without editing code.
- **Tools used:** *MATLAB, Partial Differential Equations, Computer Vision, Multi-Threading, OOP.*

Discord Bot *2022 - Present*

- Built and deployed an async event-driven Node.js application serving 1000+ users, containerised with Docker and backed by a MySQL database.
- Implemented fuzzy matching and string tokenization algorithms to improve real-time query processing, and built REST API integrations, interactive GUI components and user management tooling.
- **Tools used:** *Node.js, Docker, MySQL, Discord.js, REST APIs, Regex, String Tokenization*

Skills

Languages	Python, C++ (23), JavaScript/Node.js, SQL, MATLAB
Quantitative	PDEs, Numerical Methods, Linear Algebra, Neural Networks, Optimisation
Libraries	Pytest, Playwright, Basedpyright, FastAPI, py-spy, NumPy, PyTorch, Pandas, CUDA, Plotly
Infrastructure	Docker, Podman, Ansible, Pipelines, Kubernetes, Linux, Git, Azure Cloud, Azure DevOps, MySQL, Neo4j, uv, PM2, Bash
Tools	Jupyter Notebooks, Ollama, PowerBI, Trivy, Tree-sitter, Scalene Profiling, LaTeX